

Proposed Solution

Project proposal and implementation-oriented solution definition

Project	aiblognest	Team ID	6ab4ccd16f290be8a6b460f4
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1. Project Overview

AI BlogNest is a RESTful backend designed to streamline blog-content workflows through secure authentication, blog management, database persistence, and AI-assisted content generation and summarization. The project uses Node.js, Express, MongoDB/Mongoose, JWT, bcrypt, and Google Gemini integration.

2. Objective

Provide a structured API through which users can register and authenticate, manage blog posts, persist data, and use AI services to generate or summarize blog content.

3. Scope and Key Features

Feature	Proposed / documented capability	Primary API
Authentication	User registration, login, JWT-protected routes, and profile access	/api/auth/register, /login, /profile
Blog Management	Create, read, update, and delete blog posts	/api/blogs and /api/blogs/:id
AI Blog Generation	Generate blog content using Gemini integration	POST /api/ai/generate-blog
AI Summarization	Produce concise summaries of supplied blog content	POST /api/ai/summarize
Persistence	Store application data through MongoDB and Mongoose	Database layer
Testing / Integration	Exercise endpoints using Postman or Thunder Client	Collection / environment files

4. Technology and Resources

Area	Selected technology / resource
Backend	Node.js + Express.js
Database	MongoDB + Mongoose
Authentication & Security	JWT + bcryptjs + environment variables
AI Integration	Google Gemini via @google/generative-ai
Validation / Middleware	express-validator, cors, morgan, dotenv
API Testing	Postman / Thunder Client
Runtime Configuration	PORT, MONGO_URI, JWT_SECRET, GEMINI_API_KEY, GEMINI_MODEL

5. Expected Value

The solution consolidates authentication, content CRUD, persistence, and AI assistance behind a REST API. This is intended to reduce repeated content-workflow effort and provide a consistent backend foundation for future client applications or extensions.

Source alignment: The uploaded proposal template contains a different SmartLender project example. The completed version above replaces that example with the aiblognest project while preserving the proposal-report structure.